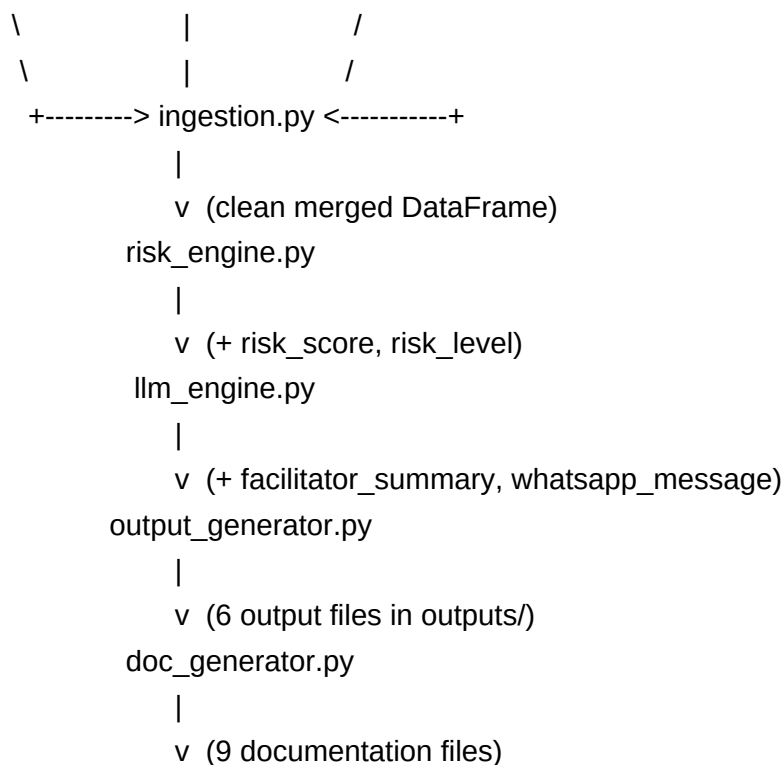


# System Architecture

## Pipeline Overview

The boon-academy-intervention pipeline processes student engagement data in five sequential stages, transforming raw CSV exports into a prioritised facilitator action list with AI-drafted WhatsApp messages. Each stage is a discrete Python module with a single public function; modules communicate only through the shared DataFrame and a run\_log dictionary - no shared mutable state, no database, no inter-process messaging.

- ingestion.py - Merges three source CSVs (student metadata, daily metrics, facilitator notes) into one clean, deduplicated DataFrame. Runs first, before any scoring or AI work.
  - risk\_engine.py - Applies a deterministic weighted formula to produce a numeric risk\_score (0 - 100) and a categorical risk\_level (CRITICAL / HIGH / MEDIUM / LOW) for every student. Pure function, no I/O, no randomness.
  - llm\_engine.py - Calls the Anthropic Claude API once per campus (not once per student) for all CRITICAL and HIGH students on that campus. Writes facilitator\_summary and whatsapp\_message columns back into the DataFrame.
  - output\_generator.py - Writes all six output artefacts: intervention\_priority\_list.xlsx, one campus dashboard per campus, whatsapp\_messages.csv, run\_log.json, dashboard.html, and intervention\_report.docx.
  - doc\_generator.py - Writes nine documentation files: analysis.md, analysis.docx, and seven technical .docx files. Called after write\_outputs() completes; uses the same df and run\_log already in scope.
- ASCII pipeline diagram: [student\_metadata.csv] [daily\_metrics.csv] [facilitator\_notes.csv]



## Component Descriptions

Each module encapsulates a single responsibility and exposes exactly one public function. This makes unit testing straightforward: each module can be tested in isolation by supplying a synthetic DataFrame and asserting on the returned DataFrame or file artefact, without spinning up the full pipeline.

- `ingestion.ingest(data_paths)` -> DataFrame - Reads CSVs with explicit dtypes to prevent ID and phone number coercion, merges on `student_id`, deduplicates by keeping the first occurrence of each duplicate, fills missing numeric activity columns with zero, and logs data quality warnings for every anomaly found.
- `risk_engine.score_risk(df)` -> DataFrame - Calculates four component scores (attendance, practice, trend, notes) using configurable weights from `config.py`, sums them into `risk_score`, assigns `risk_level` via threshold comparisons, and appends a `recommended_action` string. Accepts an optional `today` parameter for deterministic testing with `freezegun`.
- `llm_engine.enrich_with_llm(df, api_key, http_client=None)` -> tuple[DataFrame, dict] - Groups CRITICAL/HIGH students by `campus_id`, constructs a cohort prompt per campus, calls the Claude API with tool-use forcing, parses the structured response, and writes results back. Never raises an exception; uses a three-layer fallback. Returns a counts dict (`api_calls_made`, `tokens_used`, `fallbacks_triggered`).
- `output_generator.write_outputs(df, output_dir, run_log)` -> dict[str, Path] - Creates the output directory, then calls six private helpers in order. Returns a dict mapping named keys to their file paths for `main.py` logging.
- `doc_generator.write_docs(df, run_log, docs_dir)` -> dict[str, Path] - Creates the docs directory, loads `docs_content.yaml` once, then calls nine private helpers. Returns a dict mapping named keys to file paths.

## Technology Choices

Technology choices were made to minimise operational complexity for a team running daily batch jobs on a single facilitator machine, while keeping the dependency footprint auditable and the test suite reliable. Every version is pinned in `requirements.txt`.

- pandas 2.2.3 (not 3.x) - Copy-on-Write is opt-in in 2.x and mandatory in 3.x. Upgrading to 3.x would break chained assignment patterns used in `ingestion` and `risk_engine` without any functional benefit at current scale.
- openpyxl 3.1.5 (not xlswriter) - xlswriter is write-only; openpyxl supports read, write, and append. Test assertions open the generated .xlsx files and inspect cell values directly, which requires read support.
- python-docx 1.1.2 (not 1.2.0) - Version 1.2.0 has an open issue with `OxmlElement` and table border rendering. Pinned at 1.1.2 until upstream fixes are released and verified.
- PyYAML 6.0.3 - Used for loading `llm_templates.yaml` and `docs_content.yaml` via `yaml.safe_load()`. The `safe_load()` call prevents arbitrary code execution from a tampered YAML file; `yaml.load()` is never used.

- anthropic 0.103.1 - The official Anthropic Python SDK. Tool-use (structured JSON output) is used instead of text parsing, eliminating prompt-injection and markdown-stripping edge cases.
- respx 0.23.1 - HTTP mocking library for tests. The Anthropic SDK uses httpx internally; the responses library intercepts only requests-based calls and silently misses SDK calls. respx mocks at the httpx transport layer, which is exactly where the SDK sends requests.
- Jinja2 3.1.6 - Used only for dashboard.html.j2. Loaded lazily inside `_write_html_dashboard()` rather than at module import so the dependency is not paid unless the HTML output is requested.

## **Why Not n8n/Zapier or Airtable**

Several no-code and low-code platforms were evaluated against the specific requirements of the intervention pipeline. All were rejected on the same fundamental grounds: they trade debuggability, version-control, and offline reliability for ease of initial setup - a tradeoff that is wrong for a daily batch job that must be auditable by a part-time technical team at 20 campuses.

- n8n / Zapier - Workflow logic is stored as visual node graphs, not text. The pipeline cannot be code-reviewed, diffed in git, or unit-tested. Debugging a failed run requires navigating a web UI rather than reading a log file. Breaks under API rate limits with no programmatic fallback mechanism.
- Airtable - Paid tier required for API access and automation. Vendor lock-in means student PII is stored on a third-party server, raising data retention concerns. No offline operation - if the Airtable service is unavailable, the entire workflow stops.
- Google Sheets + Apps Script - Viable for very small scale (< 50 students) but becomes slow and unreliable at 300+ rows with formula dependencies. No meaningful test infrastructure; debugging requires manual inspection.
- The Python batch approach chosen here: fully version-controlled, runs offline, has 111 automated tests, produces structured logs, and costs nothing beyond the Anthropic API calls (\$200/month budget covers the projected load at 100 campuses).